

KING SAUD UNIVERSITY
COLLEGE OF COMPUTER AND INFORMATION SCIENCES
DEPARTMENT OF SOFTWARE ENGINEERING



كلية علوم الحاسب والمعلومات
قسم هندسة البرمجيات

SWE312: Software Requirements Engineering

Project

Course Code / Title: SWE 312

Assessment: Project

Semester / Year:

Submission Date:

Duration:

Deliverable

STUDENT OUTCOMES:

This phase covers/targets the following students outcomes.

Project Tasks	Relevant SOs/Clos	Marks
Functional, non-functional requirements, and design constraints	SO(2)	
Use cases, Use case description	SO(2)	
Use of SWE case tool	SO(2)	
Report organization, ethical and professional issues		
Teamwork and work distribution, Introduction		

STUDENT IDENTIFICATION:

Name: Faisal Obaid Almalki

Name: Faisal Bandar Alhammad

Name: Ibrahim Badr Alsahil

Name: Mohammed AlMusallam

Name: Abdulaziz AlAjaji

Payroll Management System

Software Requirements Specification

V3.0

2025/10/16

Prepared for
SWE 312 - Software Requirements Engineering
Spring 2025

Table of Contents

1. Introduction	7
1.1 Purpose	7
1.2 Scope	7
1.3 Definitions, Acronyms, and Abbreviations	7
1.4 References	8
1.5 Overview	8
2. General Description	8
2.1 Product Functions	8
2.2 User Characteristics	9
3. Specific Requirements	9
3.1 External Interface Requirements	9
3.2 Functional Requirements	9
3.2.1 System shall allow employees to log in using their company email and password	9
3.2.2 System shall require two-step verification by allowing the user to choose either to verify by an SMS code or verify by a biometric scan	10
3.2.3 System shall automatically generate a new timecard at the beginning of each pay period with fixed start and end dates.	10
3.2.4 System shall display the employee's current timecard, including the pay period start and end dates, daily clock-in and clock-out times upon login.	10
3.2.5 System shall allow employees to fill in their timecard by providing daily clock-in and clock-out times, project id, project name, hours worked on each project.	10
3.2.6 System shall calculate overtime hours at double the standard hourly rate for each additional hour beyond the daily 7-hour limit for hourly employees	10
3.2.7 System shall prevent modifications to the timecard after submission	10
3.2.8 System shall automatically send a weekly summary notification to hourly employees, including total hours, overtime hours, and projected earnings (calculated as the sum of daily hours up to 7 multiplied by the standard hourly rate, plus any daily hours exceeding 7 multiplied by double the standard hourly rate) before payroll runs.	11
3.2.9 System shall deposit the wages of hourly employees who have a bank deposit as their payment of choice into their bank accounts by the following Monday morning	11
3.2.10 System shall validate timecard entries by checking for missing entries.	11
3.2.11 System shall allow commissioned employees to add purchase orders by providing the following: date, product name, amount, and the customer's billing address	11
3.2.12 System shall assign a unique reference number to each purchase order.	11

- 3.2.13 System shall allow commissioned employees to edit a purchase order's details (product name or amount, or customer's billing address) by providing the purchase order reference number. 11
- 3.2.14 System shall allow commissioned employees to delete purchase orders by providing the purchase order reference number. 12
- 3.2.15 System shall calculate commission rates (10%, 15%, 25%, or 35%) for commissioned employees based on their number of sales on the second working day of each month, and add it to their base pay 12
- 3.2.16 System shall allow employees to request a year-to-date pay report by providing a specified start date 12
- 3.2.17 System shall allow employees to request a total hours worked report by providing a specified date range 13
- 3.2.18 System shall allow employees to request a vacation balances' report by providing a specified date range 13
- 3.2.19 System shall allow employees to request a remaining sick days report by providing a yearly quarter. 13
- 3.2.20 System shall allow employees to request the hours devoted to a particular project report by providing a specified date range and project id. 14
- 3.2.21 System shall validate the requested date range by checking if there exists any data in the specified date range and if the date range is a valid number. 14
- 3.2.22 System shall generate reports of various types (year-to-date pay, total hours worked, vacation balances', remaining sick days, hours devoted to a particular project) 15
- 3.2.23 System shall allow employees to export reports of various types (year-to-date pay, total hours worked, vacation balances', remaining sick days, hours devoted to a particular project) to Microsoft Excel. 15
- 3.2.24 System shall allow employees to print reports of various types (year-to-date pay, total hours worked, vacation balances', remaining sick days, hours devoted to a particular project). 15
- 3.2.25 System shall allow the payroll administrator to login to the system by providing a company email address and a password. 15
- 3.2.26 System shall allow the payroll administrator to add a new employee by providing employee name, employee national ID, employee position, and employment type (hourly employee, commissioned employee, salaried employee), base salary (salaried employees and commissioned employees) . 15
- 3.2.27 System shall allow the payroll administrator to remove an employee by providing the employee's national ID. 16
- 3.2.28 System shall allow the payroll administrator to edit an employee record by providing the employee's national ID with the edited record details: salary (for salaried and

commissioned employees), commission rate (for commissioned employees), hourly pay(for hourly employees), and tax deductions.	16
3.2.29 System shall allow the payroll administrator to add a health insurance enrollment record for an employee by providing the employee's national ID.	16
3.2.30 System shall allow the payroll administrator to remove a health insurance enrollment record for an employee by providing the employee's national ID.	16
3.2.31 System shall allow the payroll administrator to edit an employee's retirement plan contribution by providing the employee's national ID and the edited record details: the employee contribution percentage rate.	16
3.2.32 System shall allow the payroll administrator to generate high-level reports (total hours worked by a specific team, year-to-date payroll figures, and overtime trends across departments) by providing the date-range (start and end dates) and employee's national ID.	17
3.2.33 System shall require the payroll administrator to re-authenticate with an administrator token before performing sensitive actions (adding or removing employees, updating salary or tax information, and adding or removing employee benefits (health insurance and retirement contributions)).	17
3.2.34 System shall allow employees to specify their payment of choice (cash check, or bank deposit), bank details shall be provided by the employee if the bank deposit selection is chosen (Bank name and International Bank Account Number)	17
3.2.35 System shall process all employee timecards by calculating their paychecks (For salaried employees the system shall take the fixed base salary as their calculated paycheck, for commissioned employees the system shall apply (multiply) the commission rate to number of sales from the start of the second working day of the previous month till the second working day of the current month and add it to their base salary, For hourly employees, calculate daily pay by multiplying the first 7 hours by the hourly rate and any hours exceeding 7 by twice the hourly rate, then sum the daily totals for the week) applying deductions, and preparing payments every Saturday morning for hourly employees and on the second working day of each month for commissioned and salaried employees.	18
3.3 Use Cases	19
3.3.1 Use case diagram	19
3.3.2 Table of use cases	20
3.4 Use Cases Description	21
3.4.1 Use Case #1	21
3.4.2 Use Case #2	23
3.4.3 Use Case #3	24
3.4.4 Use Case #4	25
3.4.5 Use Case #5	26

3.6 Non-Functional Requirements	27
3.6.1 Performance	27
3.6.2 Reliability	27
3.6.3 Availability	27
3.6.4 Security	27
3.6.5 Maintainability	27
3.6.6 Portability	27
3.6.6.1 System shall run on mobile phones	27
3.7 Design Constraints	27
3.8 Other Requirements	27
4. Analysis Models	28
4.1 Sequence Diagrams and Mockup Screens	28
4.2 Activity Diagrams	32
4.2.1 Add employee	32
4.2.1 Remove employee	33
5 Classes / Objects	34
5.1 Class diagram	34
5.2 Employee	35
5.3 HourlyEmployee	35
5.4 SalariedEmployee	35
5.5 CommissionedEmployee	35
5.6 PurchaseOrder	36

1. Introduction

1.1 Purpose

The purpose of the *PayThemAll Payroll Management System* is to replace the company's old and unreliable payroll platform with a modern, secure, and efficient system. The new system aims to automate payroll operations, improve reliability, enhance security, and provide faster and more accurate processing for all employees. This SRS is intended for the system designers who will use it to design the payroll management system.

1.2 Scope

The *PayThemAll Payroll Management System* is designed to automate and manage payroll operations for *PayThemAll*.

The system aims to replace the unreliable legacy payroll platform with a faster, more secure, and accurate solution. It supports different employee types (hourly, salaried, and commissioned) and ensures timely and error-free salary processing.

1.3 Definitions, Acronyms, and Abbreviations

1.3.1 Bank System: Any external banking system used to process and deposit employee payments electronically

1.3.2 Employee: A person who works for PayThemAll, Inc., including hourly, salaried, and commissioned employees.

1.3.3 Payroll Administrator: The person responsible for maintaining employees and employee information in the system.

1.3.4 Work Management Database (WMD): The company's legacy database that stores all project charge numbers. The payroll system can read from this database but cannot modify it

1.3.5 System Clock: The internal system clock that keeps track of time. The internal clock will automatically run the payroll at the appropriate times.

1.3.6 Pay Period: The time duration over which an employee's pay is calculated (e.g., weekly or monthly).

1.3.7 Paycheck: A record showing the total amount paid to an employee for a specific pay period.

1.3.8 Payment Method: The way an employee receives payment — either by direct deposit or by collecting a check at the office.

1.3.9 Timecard: A digital record of hours worked by an employee during a pay period

1.3.10 Purchase Order: A sales record created by commissioned employees that includes details such as date, product, amount, and customer billing information.

1.3.11 Hourly Employee: An employee paid based on the number of hours worked.

1.3.12 Salaried Employee: An employee who receives a fixed monthly salary regardless of hours worked.

1.3.13 Commissioned Employee: An employee who earns both a base salary and commission based on sales performance.

1.3.14 Administrator Token: A digital security credential used to re-authenticate payroll administrators before performing sensitive system operations(Assuming this is what it means).

1.4 References

1.5 Overview

The rest of the SRS will contain a section for the general description of the Payroll Management System which includes the product functions and user characteristics, Section 3 will be about specific requirements where it will detail external interface requirements, functional requirements, use cases, use cases description, non-functional requirements, and design constraints.

2. General Description

2.1 Product Functions

2.1.1 Logging in: Allows employees and administrators to securely access the system using valid credentials.

2.1.2 Managing timecards: Enables employees to view, record, edit, and submit their timecards for each pay period.

2.1.3 Request reports: Allows employees to request detailed reports based on data type, date range.

2.1.4 Request high-level reports: Allows administrators to request detailed reports based on data type, date range, and report level

2.1.5 Creating purchase orders: Allows commissioned employees to create new purchase orders.

2.1.6 Editing or removing purchase orders: Allows commissioned employees to modify or delete existing purchase orders using their reference numbers.

2.1.7 Adding employees: Enables payroll administrators to add new employees to the system.

2.1.8 Managing employee records: Allows payroll administrators to view and update employee information.

2.1.9 Removing employees: Enables payroll administrators to remove employees from the system when necessary.

2.1.10 Running payroll: Automates salary calculations, deductions, and payment processing for all employees during each pay cycle.

2.2 User Characteristics

Hourly and Salaried employees: The functions that Hourly and Salaried can do are log into the system, manage timecards, and request reports.

Commissioned employees: The functions that Hourly and Salaried can do are log into the system, manage timecards, request reports, create purchase orders, edit purchase orders, and remove purchase orders

Payroll Administrator: The functions that payroll administrator can do are log into the system, request high-level reports, add new employees, manage and update employee records, and remove employees from the system.

All users should have the technical ability to use mobile phones or tablets.

Comment: we are defining user Characteristics by product function use and technical expertise.

3. Specific Requirements

3.1 External Interface Requirements

3.1.1 System shall interact with SMS and Biometric providers to ensure two step verification.

3.1.2 System shall be able to fetch and read from the legacy work management database

3.1.3 System shall integrate with external banking systems to automatically transfer employee payroll deposits.

3.1.4 System shall allow salaried employees to export their recorded hours into external project dashboards.

3.1.5 System shall be able to interact with the company's HR (Human Resources) management system to fetch remaining sick days and vacation balances of employees.

3.2 Functional Requirements

3.2.1 System shall allow employees to log in using their company email and password

3.2.1.1 Comments: none

3.2.1.2 Inputs: company email and password

3.2.1.3 Outputs: none

3.2.2 System shall require two-step verification by allowing the user to choose either to verify by an SMS code or verify by a biometric scan

3.2.2.1 Comments: none

3.2.2.2 Inputs: SMS verification code or biometric scan

3.2.2.3 Outputs: none

3.2.3 System shall automatically generate a new timecard at the beginning of each pay period with fixed start and end dates.

3.2.3.1 Comments: none

3.2.3.2 Inputs: none

3.2.3.3 Outputs: new timecard created with predefined start and end dates

3.2.4 System shall display the employee's current timecard, including the pay period start and end dates, daily clock-in and clock-out times upon login.

3.2.4.1 Comments: There is an ambiguity in the system description, it states that the timecard will be displayed to the employee upon login, but the details of the timecard weren't exactly stated so I assumed the following logical information (pay period start and end dates, daily clock-in and clock-out times).

3.2.4.2 Inputs: none

3.2.4.3 Outputs: timecard displayed on dashboard

3.2.5 System shall allow employees to fill in their timecard by providing daily clock-in and clock-out times, project id, project name, hours worked on each project.

3.2.5.1 Comments: This requirement is ambiguous since the details asked to be provided by the system are not exactly clear, therefore I've assumed the appropriate details for a timecard 3.2.5.2 Inputs: daily clock-in and clock-out times, project id, project name, hours worked on each project.

3.2.5.3 Outputs: none

3.2.6 System shall calculate overtime hours at double the standard hourly rate for each additional hour beyond the daily 7-hour limit for hourly employees

3.2.6.1 Comments: none

3.2.6.2 Inputs: total daily working hours

3.2.6.3 Outputs: none

3.2.7 System shall prevent modifications to the timecard after submission

3.2.7.1 Comments: none

3.2.7.2 Inputs: none

3.2.7.3 Outputs: none

3.2.8 System shall automatically send a weekly summary notification to hourly employees, including total hours, overtime hours, and projected earnings (calculated as the sum of daily hours up to 7 multiplied by the standard hourly rate, plus any daily hours exceeding 7 multiplied by double the standard hourly rate) before payroll runs.

3.2.8.1 Comments: Here projected earnings we assume is to show the employee their expected paycheck at the end of the month and also to distinguish between regular work hours and overtime work hours

3.2.8.2 Inputs: none

3.2.8.3 Outputs: weekly summary notification

3.2.9 System shall deposit the wages of hourly employees who have a bank deposit as their payment of choice into their bank accounts by the following Monday morning

3.2.9.1 Comments: The payment of choice wasn't specified but depending on context from the last two paragraphs, the employee can either have a bank deposit or cash check collection as their payment of choice, so to be clear and specific, we noted this in the requirement.

3.2.9.2 Inputs: none

3.2.9.3 Outputs: none

3.2.10 System shall validate timecard entries by checking for missing entries.

3.2.10.1 Comments: The description stated that Ali would get notified if there are any missing entries', We have deduced from this sentence that the system shall validate all employees' timecards for missing entries and notify them to complete their timecards.

3.2.10.2 Inputs: none

3.2.10.3 Outputs: error alert notification

3.2.11 System shall allow commissioned employees to add purchase orders by providing the following: date, product name, amount, and the customer's billing address

3.2.11.1 Comments: none

3.2.11.2 Inputs: purchase order details (date, product name, amount, billing address)

3.2.11.3 Outputs: none

3.2.12 System shall assign a unique reference number to each purchase order.

3.2.12.1 Comments: none

3.2.12.2 Inputs: none

3.2.12.3 Outputs: generated reference number

3.2.13 System shall allow commissioned employees to edit a purchase order's details (product name or amount, or customer's billing address) by providing the purchase order reference number.

3.2.13.1 Comments: The editable fields were not stated in the description, so we assumed that the purchase order editable details are product name or amount, or customer's billing address; however, the date shall not be editable.

3.2.13.2 Inputs: purchase order reference number, (product name or amount or customer's billing address)

3.2.13.3 Outputs: none

3.2.14 System shall allow commissioned employees to delete purchase orders by providing the purchase order reference number.

3.2.14.1 Comments: none

3.2.14.2 Inputs: purchase order reference number

3.2.14.3 Outputs: none

3.2.15 System shall calculate commission rates (10%, 15%, 25%, or 35%) for commissioned employees based on their number of sales on the second working day of each month, and add it to their base pay

3.2.15.1 Comments: There is an ambiguity, because the text does not mention the criteria based on which an employee would get a percentage of 10,15,25, or 35, and also the payroll administrator has the right to set the commission rate, so logically the commission rate is directly linked with how many sales the employee has recorded. So here is the following criteria on which the system shall calculate the commission rate.

Number of Sales	Commission rate
0-5000 Saudi Arabian Riyals	10%
5000-10,000 Saudi Arabian Riyals	15%
10,000 – 15,000 Saudi Arabian Riyals	25%
15,000 – 20,000 Saudi Arabian Riyals	35%

3.2.15.2 Inputs: None.

3.2.15.3 Outputs: Commission rate

3.2.16 System shall allow employees to request a year-to-date pay report by providing a specified start date

3.2.16.1 Comments: There is an ambiguity whether we start calculating from the beginning of the current year or start calculating a year from a provided start date from the user, We assumed that the user will provide a start date and the system will calculate a year's pay depending on it.

3.2.16.2 Inputs: start date

3.2.16.3 Outputs: report

Report type	YEAR TO DATE PAY REPORT
Employee Name	Faisal Almalki
Employee ID	12345
period	2025-03-03 to 2025-07-04
Year-to-Date Pay	50,000\$
Total	50,000\$

3.2.17 System shall allow employees to request a total hours worked report by providing a specified date range

3.2.17.1 Comments: none

3.2.17.2 Inputs: date range (start date, end date)

3.2.17.3 Outputs: report

Report type	total hours worked
Employee name	Ibrahim Alsahil
Employee ID	221133
period	2025-06-01 to 2025-10-03
Total hours worked	720

3.2.18 System shall allow employees to request a vacation balances' report by providing a specified date range

3.2.18.1 Comments: To remove ambiguities in this requirement, we have assumed that the company has an employee HR (Human Resources) management system that contains vacation balances and remaining sick days for each employee. Thus, we deduced the external interface requirement 3.1.5

3.2.18.2 Inputs: date range (start date, end date)

3.2.18.3 Outputs: report

Report type	Vacation Balances
Employee name	Faisal Alhammad
Employee ID	459398
period	2025-01-01 to 2025-12-3
Vacation Days Allocated	30
Vacation Days Used	18
Vacation Days Remaining	12

3.2.19 System shall allow employees to request a remaining sick days report by providing a yearly quarter.

3.2.19.1 Comments: To remove ambiguities in this requirement, we have assumed that the company has an employee HR (Human Resources) management system that contains vacation

balances and remaining sick days for each employee. Thus, we deduced the external interface requirement 3.1.5

3.2.19.2 Inputs: fixed quarter of a year (Q1: from Jan to March, Q2: from April to June, Q3: from July to September, Q4: from October to December)

3.2.19.3 Outputs: report

Report type	Remaining Sick Days Report
Employee name	Zaid Alameer
Employee ID	768396
Quarter	Q1(2025-01-01 to 2025-03-31)
Total Sick Days Allocated	10
Sick Days Used	3
Sick Days Remaining	7

3.2.20 System shall allow employees to request the hours devoted to a particular project report by providing a specified date range and project id.

3.2.20.1 Comments: none

3.2.20.2 Inputs: date range (start date, end date), project id

3.2.20.3 Outputs: report

Report type	Hours Devoted to Project Report
Employee name	Ahmed Atiar
Employee ID	768396
Project name	HR System Enhancement
Project ID	PRJ-2025-042
period	2025-05-01 to 2025-09-30
Total Hours Worked on Project	290

3.2.21 System shall validate the requested date range by checking if there exists any data in the specified date range and if the date range is a valid number.

3.2.21.1 Comments: There is an ambiguity in the system description, the customer asks to validate the request by checking it is unsupported but what does unsupported exactly mean? We

concluded that we are referring to data existence in this date range and also if the date range is a valid number.

3.2.21.2 Inputs: date range (start date, end date)

3.2.21.3 Outputs: validation message or corrected date suggestion

3.2.22 System shall generate reports of various types (year-to-date pay, total hours worked, vacation balances', remaining sick days, hours devoted to a particular project)

3.2.21.1 Comments: none

3.2.21.2 Inputs: none

3.2.21.3 Outputs: exported report

3.2.23 System shall allow employees to export reports of various types (year-to-date pay, total hours worked, vacation balances', remaining sick days, hours devoted to a particular project) to Microsoft Excel.

3.2.23.1 Comments: none

3.2.23.2 Inputs: none

3.2.23.3 Outputs: Excel Sheet.

3.2.24 System shall allow employees to print reports of various types (year-to-date pay, total hours worked, vacation balances', remaining sick days, hours devoted to a particular project).

3.2.24.1 Comments: none.

3.2.24.2 Inputs: none.

3.2.24.3 Outputs: none.

3.2.25 System shall allow the payroll administrator to login to the system by providing a company email address and a password.

3.2.25.1 Comments: This was briefly mentioned in the description and is ambiguous, what details will the administrator provide to login to the system? We assumed that the admin login details are as mentioned in 3.2.23.2

3.2.25.2 Inputs: Company email address, password

3.2.25.3 Outputs: none

3.2.26 System shall allow the payroll administrator to add a new employee by providing employee name, employee national ID, employee position, and employment type (hourly employee, commissioned employee, salaried employee), base salary (salaried employees and commissioned employees) .

3.2.26.1 Comments: I assume the mandatory Information of each employee are name, ID, position, and employment type

3.2.26.2 Inputs: new employee details (name, ID, position, and employment type, base salary for salaried and commissioned employees')

3.2.26.3 Outputs: new employee record created

3.2.27 System shall allow the payroll administrator to remove an employee by providing the employee's national ID.

3.2.27.1 Comments: I assume the unique and mandatory field to remove an employee is their national ID.

3.2.27.2 Inputs: national ID

3.2.27.3 Outputs: none

3.2.28 System shall allow the payroll administrator to edit an employee record by providing the employee's national ID with the edited record details: salary (for salaried and commissioned employees), commission rate (for commissioned employees), hourly pay (for hourly employees), and tax deductions.

3.2.28.1 Comments: none

3.2.28.2 Inputs: salary (for salaried and commissioned employees), commission rate (for commissioned employees'), hourly pay (for hourly employees'), and tax deductions.

3.2.28.3 Outputs: none

3.2.29 System shall allow the payroll administrator to add a health insurance enrollment record for an employee by providing the employee's national ID.

3.2.29.1 Comments: In the system description this was vaguely mentioned and the word ("managed") was used which makes the requirement un-testable, hence we deduced the following requirements 3.2.27, 3.2.28, 3.2.29.

3.2.29.2 Inputs: employee's national ID, health insurance details (provider name, grade (silver, gold, platinum), subscription price, start date, end date)

3.2.29.3 Outputs: none

3.2.30 System shall allow the payroll administrator to remove a health insurance enrollment record for an employee by providing the employee's national ID.

3.2.30.1 Comments: In the system description this was vaguely mentioned and the word ("managed") was used which makes the requirement un-testable, hence we deduced the following requirements 3.2.27, 3.2.28, 3.2.29.

3.2.30.2 Inputs: employee's national ID.

3.2.30.3 Outputs: none

3.2.31 System shall allow the payroll administrator to edit an employee's retirement plan contribution by providing the employee's national ID and the edited record details: the employee contribution percentage rate.

3.2.31.1 Comments: In the system description this was vaguely mentioned and the word ("managed") was used which makes the requirement un-testable, hence we deduced the following requirements 3.2.27, 3.2.28, 3.2.29. In this requirement we assumed that with the creation of each employee the default employee contribution percentage is 5% and payroll administrators shall be able to edit this percentage rate for each employee.

3.2.31.2 Inputs: employee's national ID, employee contribution percentage rate.

3.2.31.3 Outputs: none

3.2.32 System shall allow the payroll administrator to generate high-level reports (total hours worked by a specific team, year-to-date payroll figures, and overtime trends across departments) by providing the date-range (start and end dates) and employee's national ID.

3.2.32.1 Comments: We assume that the term ("Review") here is outside the system boundary, hence the payroll administrator will generate a report for each customer and "review" the total trends. The design for year-to-date and total hours shall be the same, we have only attached the design for the overtime employee report.

3.2.32.2 Inputs: Employee's National ID, date-range

3.2.32.3 Outputs: high-level report

Report type	Overtime Hours Report
Employee name	Norah Almutairi
Employee ID	324574
period	2025-06-01 to 2025-09-30
Regular Hours Worked	640
Overtime Hours Worked	58
Overtime Pay Rate	1.5x
Total Overtime Pay	4,000\$
Total Pay Including Overtime	28,000\$

3.2.33 System shall require the payroll administrator to re-authenticate with an administrator token before performing sensitive actions (adding or removing employees, updating salary or tax information, and adding or removing employee benefits (health insurance and retirement contributions)).

3.2.33.1 Comments: Sensitive actions include those that affect payroll data integrity or employee records, I have limited them on actions (adding or removing employees, updating salary or tax information, and modifying employee benefits)

3.2.33.2 Inputs: admin token

3.2.33.3 Outputs: access granted to perform sensitive actions

3.2.34 System shall allow employees to specify their payment of choice (cash check, or bank deposit), bank details shall be provided by the employee if the bank deposit selection is chosen (Bank name and International Bank Account Number)

3.2.34.1 Comments: Here no details are being provided by the employee in the system description, let's say employee chooses bank deposit how will the system deposit money to an undefined bank account. Thus we assumed the following inputs in 3.2.31.2

3.2.34.2 Inputs: (Bank name and International Bank Account Number)

3.2.34.3 Outputs: none

3.2.35 System shall process all employee timecards by calculating their paychecks (For salaried employees the system shall take the fixed base salary as their calculated paycheck, for commissioned employees the system shall apply (multiply) the commission rate to number of sales from the start of the second working day of the previous month till the second working day of the current month and add it to their base salary, For hourly employees, calculate daily pay by multiplying the first 7 hours by the hourly rate and any hours exceeding 7 by twice the hourly rate, then sum the daily totals for the week) applying deductions, and preparing payments every Saturday morning for hourly employees and on the second working day of each month for commissioned and salaried employees.

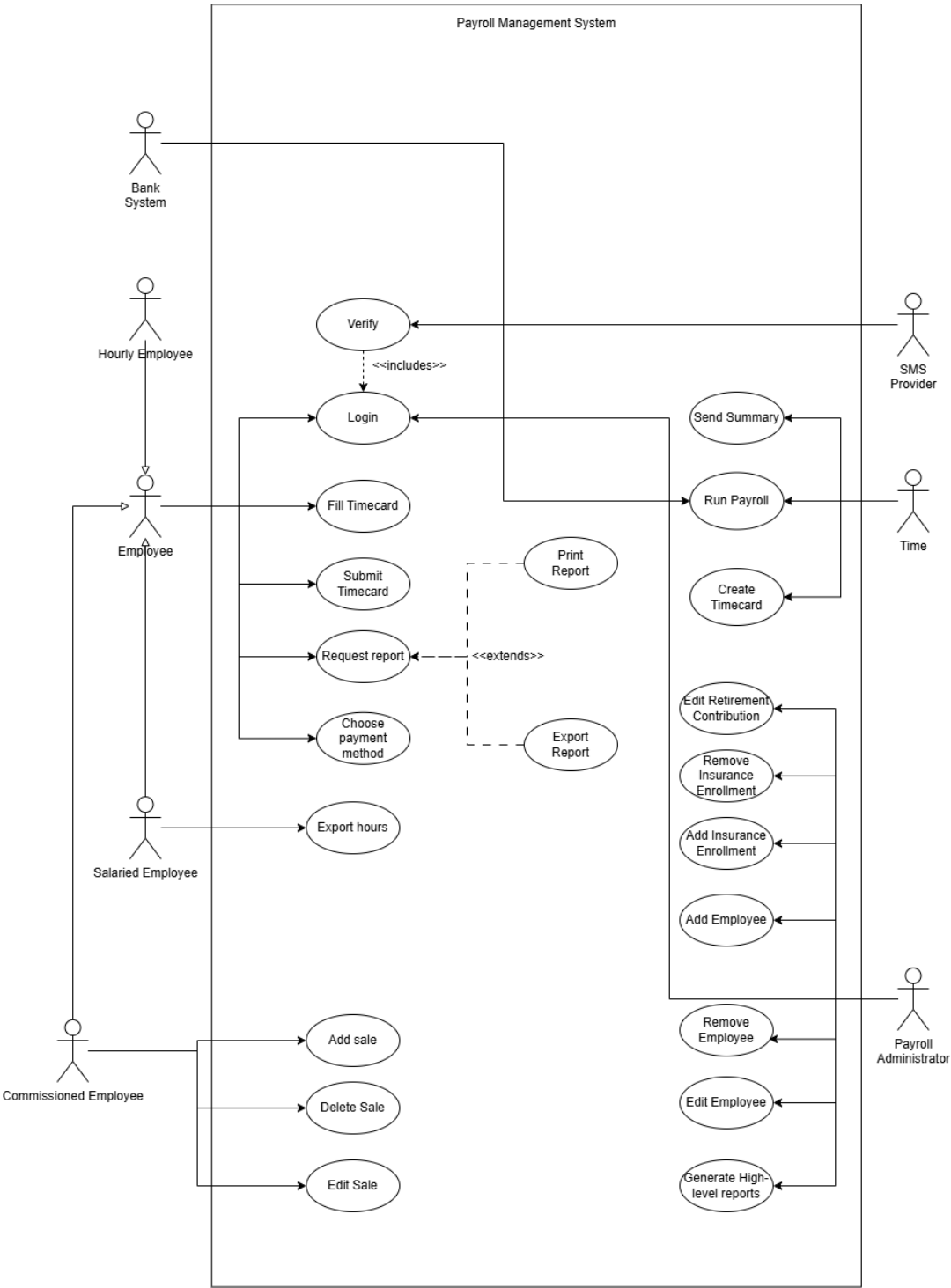
3.2.35.1 Comments: In the last paragraph of the system description this was summarized quickly without exactly stating who exactly is getting processed on Saturday and who is getting processed on the second working day of each month, but based on the context of previous paragraphs we concluded the following requirement.

3.2.35.2 Inputs: none

3.2.35.3 Outputs: none

3.3 Use Cases

3.3.1 Use case diagram



3.3.2 Table of use cases

Use case	Actor	Relationships
UC1: Login	Employee, Administrator	-
UC2: Verify	Employee	Includes UC1
UC3: Fill Timecard	Employee	-
UC4: Submit Timecard	Employee	-
UC5: Request Report	Employee	-
UC6: Print	Employee	Extends UC5
UC7: Export	Employee	Extends UC5
UC8: Choose Payment Method	Employee	-
UC9: Export Hours	Salaried Employee	-
UC10: Add Sale	Commissioned Employee	-
UC11: Delete Sale	Commissioned Employee	-
UC12: Edit Sale	Commissioned Employee	-
UC13: Create Timecard	Time	-
UC14: Run Payroll	Time	-
UC15: Send Summary	Time	-
UC16: Add Employee	Administrator	-
UC17: Remove Employee	Administrator	-
UC18: Edit Employee	Administrator	-
UC19: Generate High-Level Reports	Administrator	-
UC20: Add Insurance Enrollment	Administrator	-
UC21: Remove Insurance Enrollment	Administrator	-
UC22: Edit Retirement Contribution	Administrator	-

3.4 Use Cases Description

3.4.1 Use Case #1

Use Case Description	
System: Payroll Management System	
Use Case name: Add Employee	
Primary actor: Payroll Administrator	Other actors: None
Stakeholders: Employee (hourly, commissioned, salaried)	
Description: This use case allows the payroll administrator to add new employee by entering required personal and employment information into the system.	
Relationships ▪Includes: none ▪Extends: none	
Input: employee name, employee national ID, employee position, employment type, base salary (just for salaried employee and commissioned employee)	
Pre-conditions: The payroll administrator is logged in and authenticated with an administrator token	
Steps:	.
Actor	System
1.The administrator asks the system to add a new employee	2- The system asks for employee details (Employee name, ID, Position, employment type)
3.The administrator provides the required information (Employee name, ID, Position, employment type)	4.The System checks that the employee information does not exist in the system
	5.The system saves the new employee information
	6.The system displays a success message
Alternative and exceptional flows: 3a: The administrator left one or more required fields (employee name, ID, position, and employment type) 3a1. The system displays error messages then ask the admin return to step 2 of the basic flow 3a2 The system asks the administrator whether they want to continue the employee-addition process. If the administrator chooses “Yes,” the system returns to Step 2 of the basic flow. If the administrator chooses “No,” the use case terminates gracefully 3b: The administrator entered in the employment type field (Salaried or Commissioned)	

3b1. The system asks for base salary
3b2. The administrator provided the base salary
3b3. The flow returns to step 4 of the basic flow
4a: There is an employee registered in the system with this information entered
4a1. The system displays error messages
4a2 The system asks the administrator whether they want to continue the employee-addition process. If the administrator chooses “Yes,” the system returns to Step 2 of the basic flow. If the administrator chooses “No,” the use case terminates gracefully

Post-conditions: The employee is added

employee information: Employee name, ID, Position, type, base salary for (Salaried or Commissioned)

3.4.2 Use Case #2

Use Case Description	
System: Payroll Management System	
Use Case name: Remove Employee	
Primary actor: Payroll Administrator	Other actors: none
Stakeholders: Employee (hourly, commissioned, salaried)	
Description: This use case allows the payroll administrator to remove an existing employee's record from the payroll system.	
Relationships ▪Includes: none ▪Extends: none	
Input: National ID	
Pre-conditions: The payroll administrator is logged in and authenticated with an administrator token	
Steps:	.
Actor	System
1.The administrator asks the system to remove an employee 3.The administrator provides the national ID	2. The system asks for the employee's national ID 4. The system checks if the employee national ID exists in the system 5. The system removes the employee 6. The system displays a success message
Alternative and exceptional flows: 3a: The administrator entered an invalid value in the National ID field 3a1. The system displays error messages 3a2. The system asks the administrator whether they want to continue the employee-removal process. If the administrator chooses “Yes,” the system returns to Step 2 of the basic flow. If the administrator chooses “No,” the use case terminates gracefully 4a: There is no employee registered in the system with this National ID entered 4a1. The system displays error messages 4a2. The system asks the administrator whether they want to continue the employee-removal process. If the administrator chooses “Yes,” the system returns to Step 2 of the basic flow. If the administrator chooses “No,” the use case terminates gracefully	
Post-conditions: The employee is removed	

3.4.3 Use Case #3

Use Case Description	
System: Payroll Management System	
Use Case name: Add Purchase Order	
Primary actor: Employee (Commissioned)	Other actors: none
Stakeholders: none	
Description: This use case allows a commissioned employee to record a purchase order by entering details (date, product name, amount, customer billing address).	
Relationships ▪Includes: none ▪Extends: none	
Input: date, product name, amount, customer billing address	
Pre-conditions: The employee logged in	
Steps:	.
Actor	System
1.The employee asks the system to add a new purchase order 3.The employee enters order details (date, product name, amount and customer billing address)	2.The system asks the employee to provide order details (date, product name, amount and customer billing address) 4.The system validates* the order details (date, product name, amount and customer billing address) 5.The system creates the purchase order and assigns a unique order number 6.The system displays a confirmation message
Alternative and exceptional flows: 4a: If any field is missing (Date, product name, amount, customer billing address) 4a1.The system highlights the missing fields 4a2.The system asks the employee to complete them.	
Post-conditions: purchase order record is created	

***validates:** In this context the system shall validate by checking if the entered date range is in valid date format (non negative numbers and months between 0 to 12 and days between 0 to 31 depending on the months since some contain 28 or 29 or 30 or 31), the amount by checking if it's a non-negative number, customer billing address by checking if its non-empty

3.4.4 Use Case #4

Use Case Description	
System: Payroll Management System	
Use Case name: Delete Purchase Order	
Primary actor: Employee (Commissioned)	Other actors: none
Stakeholders: none	
Description: This use case allows a commissioned employee to delete an existing purchase order using its unique order number.	
Relationships ▪Includes: none ▪Extends: none	
Input: purchase order reference number	
Pre-conditions: The employee logged in	
Steps:	.
Actor	System
1.The employee asks the system to delete a purchase order 3.The employee provides the reference number	2.The system asks the employee to enter reference number 4.The system checks if order exists 5.The system deletes the purchase order 6. The system displays success message
Alternative and exceptional flows: 3a: If employee enters letter or character 3a1.The system displays error messages then returns to step 2 4a: If the order reference number does not exist 4a1.The system displays an error message and returns to Step 2.	
Post-conditions: The purchased order is deleted	

3.4.5 Use Case #5

Use Case Description	
System: Payroll Management System	
Use Case name: Send Summary	
Primary actor: Time	Other actors:
Stakeholders: Hourly worker	
Description: This use case describes how the system automatically generates and sends a summary report to hourly workers before payroll, using available data such as hours worked, sales, or payroll information	
Relationships ▪Includes: ▪Extends:	
Input: Sale Reference ID	
Pre-conditions:	
Steps:	.
Actor	System
1-Time triggers the summary generation before payroll runs.	2-The system generates the summary report using available data. (hours worked, sales, or payroll data at scheduled intervals) 3-The system sends the summary automatically to all hourly workers.
Alternative and exceptional flows:	
Post-conditions: The summary is generated and sent to the hourly employee	

3.6 Non-Functional Requirements

3.6.1 Performance

- 3.6.1.1 System shall support at least **9000** users.
- 3.6.1.2 Average system response time should be less than 3 seconds.
- 3.6.1.3 System shall be operational in less than 30 seconds.

3.6.2 Reliability

- 3.6.2.1 System shall calculate paychecks with accuracy of 2 decimal points.
- 3.6.2.1 System shall create backup copies of payroll data in secure cloud storage daily.

3.6.3 Availability

- 3.6.3.1 System shall remain accessible during payroll processing periods.
- 3.6.3.2 System downtime shall not exceed **1 hour per month**.

3.6.4 Security

none

3.6.5 Maintainability

none

3.6.6 Portability

- 3.6.6.1 System shall run on mobile phones
- 3.6.6.2 System shall run on tablets

3.7 Design Constraints

- 3.7.1 The backup data shall be stored in a cloud storage

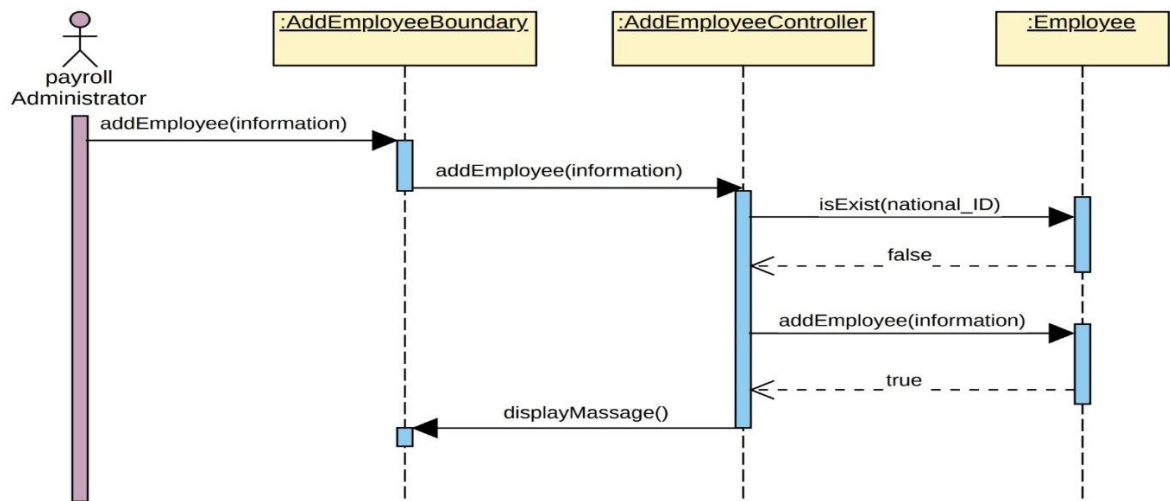
3.8 Other Requirements

none

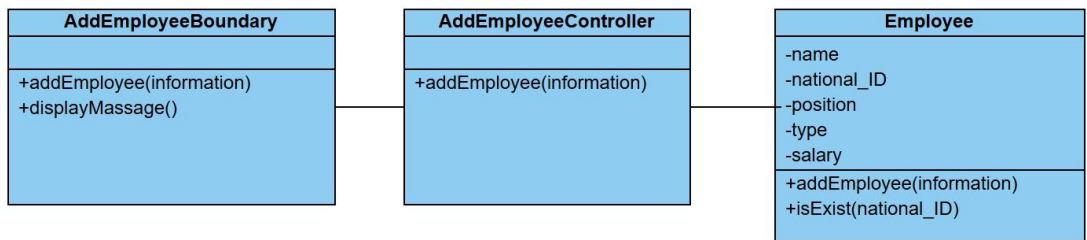
4. Analysis Models

4.1 Sequence Diagrams and Mockup Screens

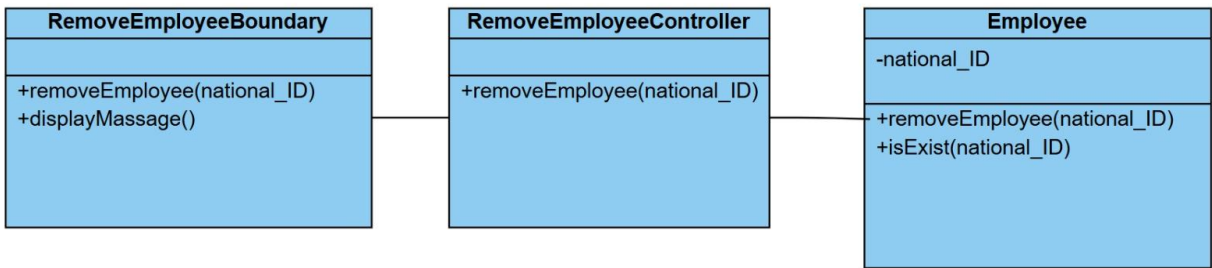
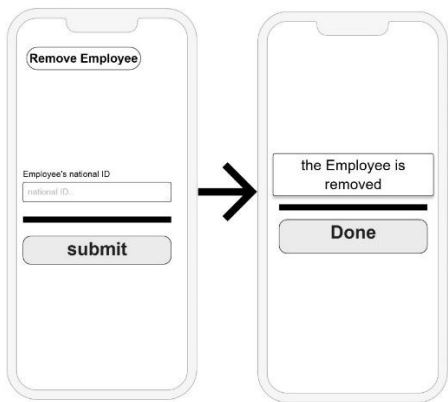
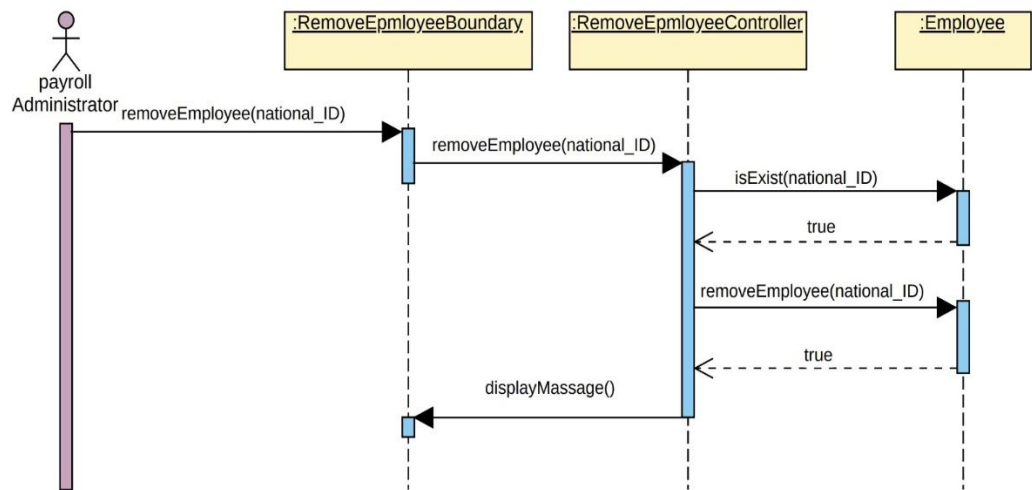
4.1.1 Add Employee Use Case



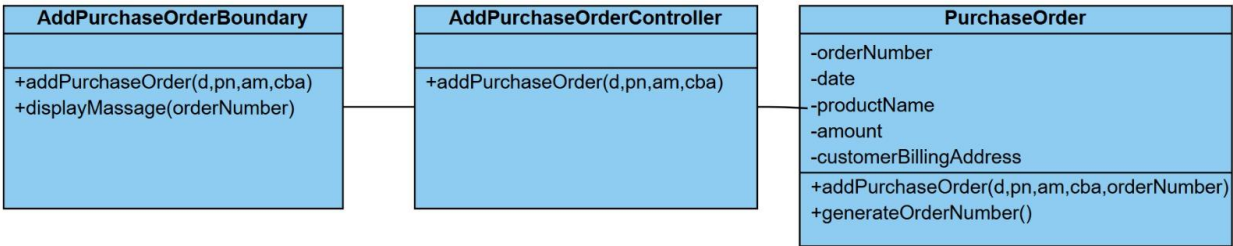
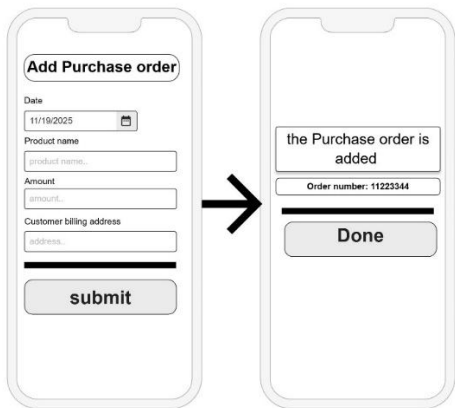
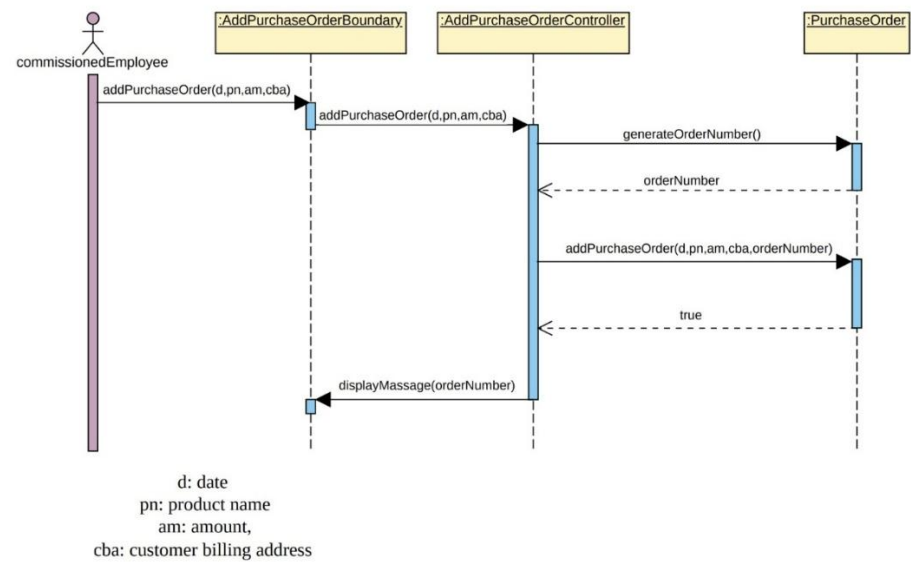
information: : employee name, employee national ID, employee position, employment type, base salary (just for salaried employee and commissioned employee)



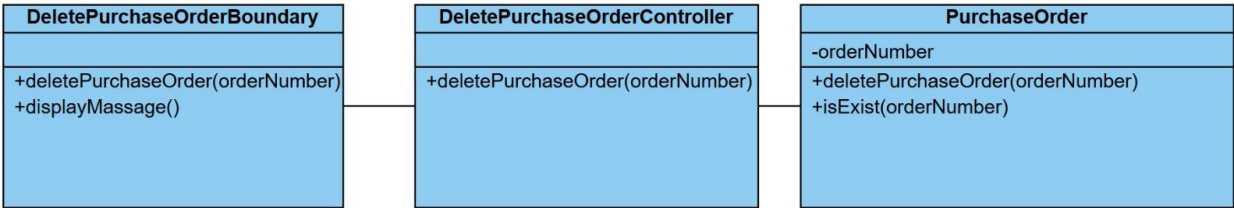
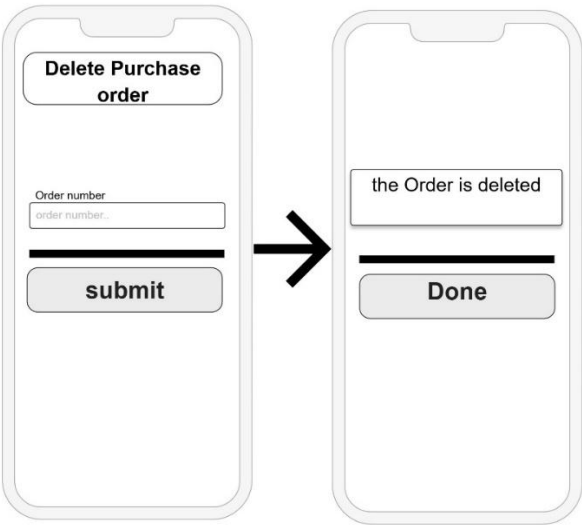
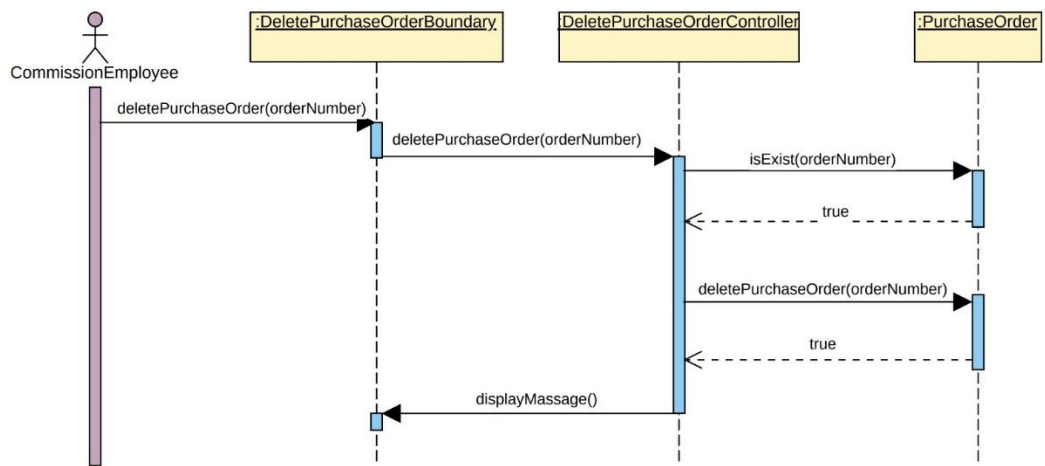
4.1.2 Remove Employee Use Case



4.1.3 Add Purchase Order Use Case

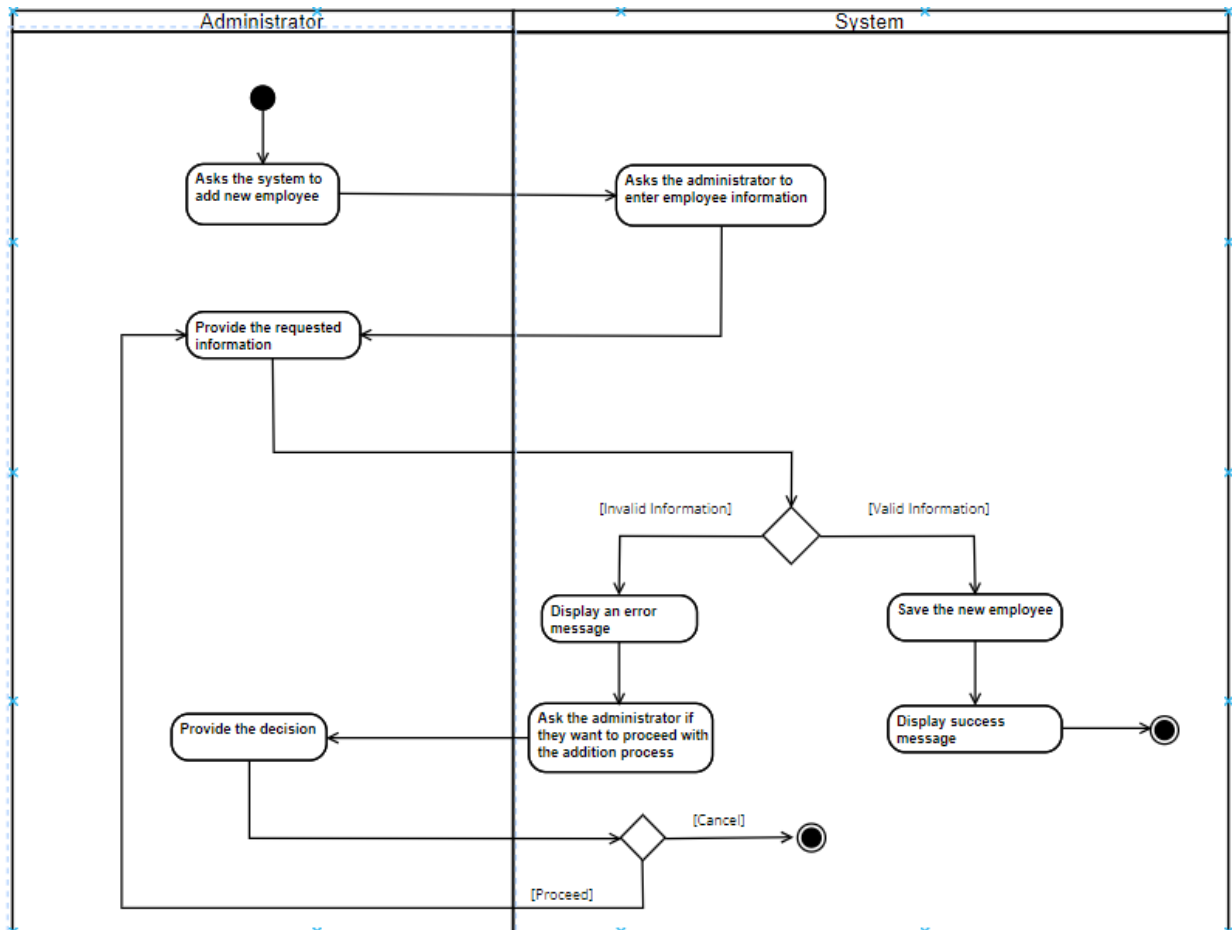


4.1.4 Delete Purchase Order Use Case



4.2 Activity Diagrams

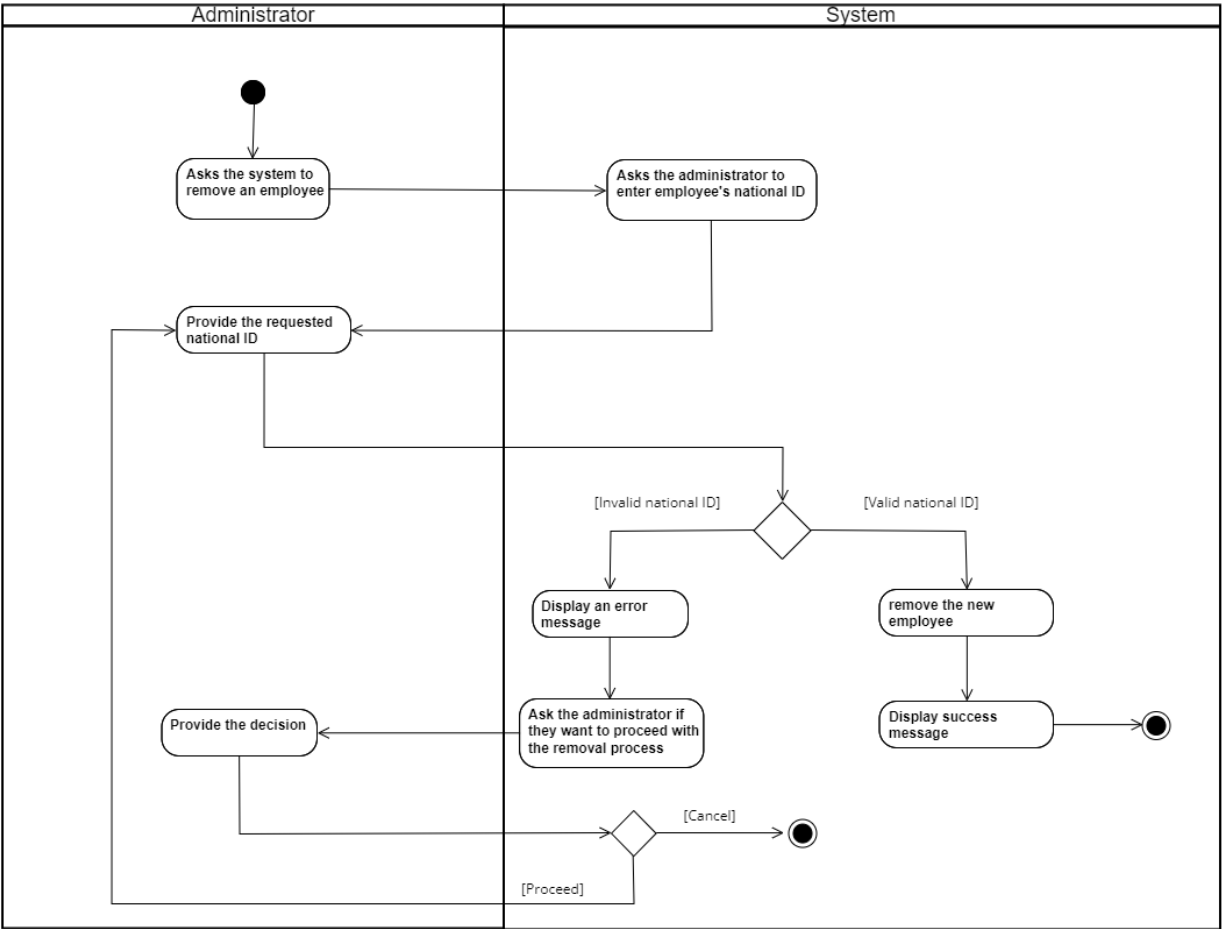
4.2.1 Add employee



* employee information: employee name, employee national ID, employee position, employment type, base salary (just for salaried employee and commissioned employee)

...

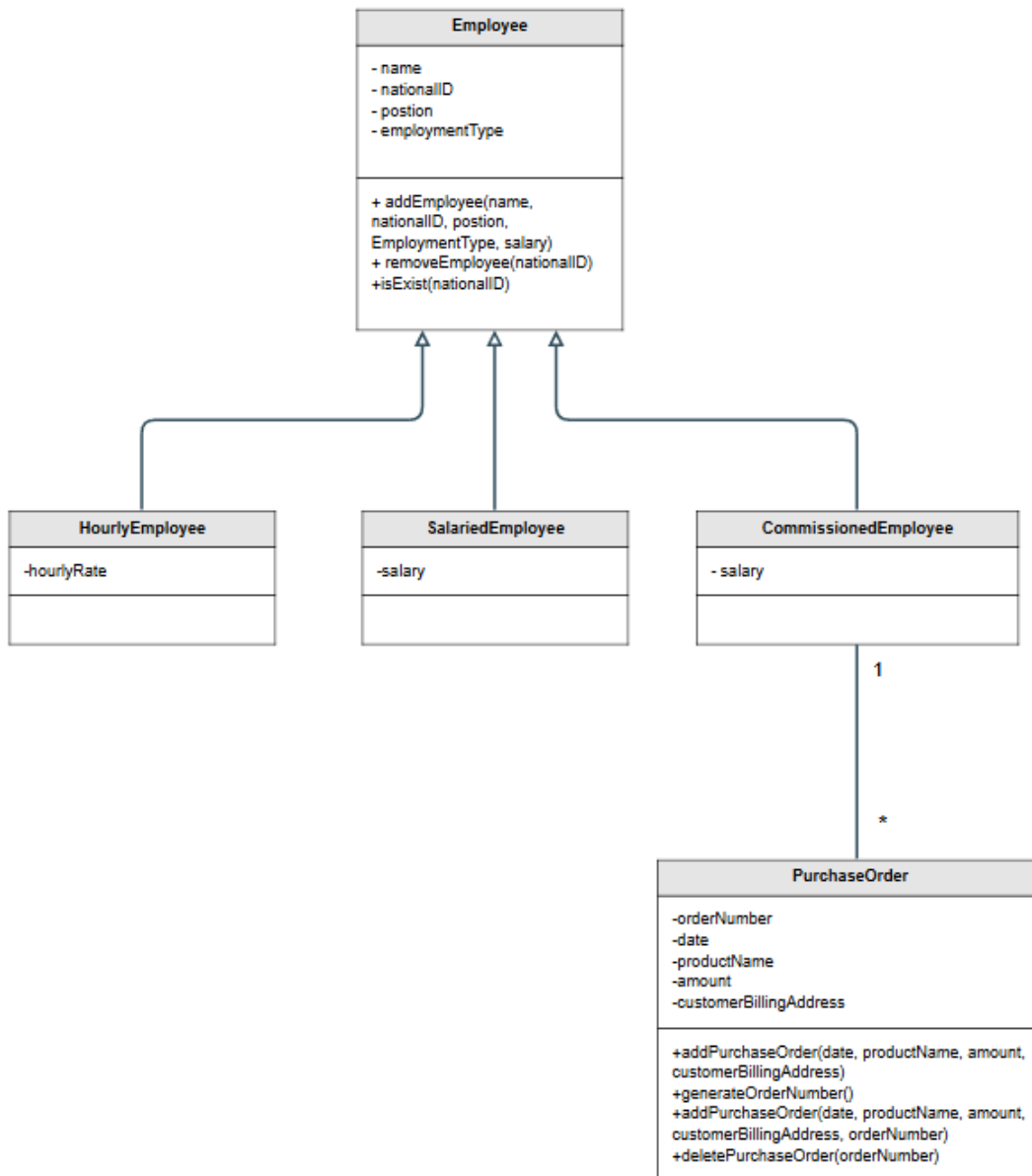
4.2.1 Remove employee



...

5 Classes / Objects

5.1 Class diagram



5.2 Employee

5.2.1 Attributes

- 5.2.1.1 name
- 5.2.1.2 nationalID
- 5.2.1.3 position
- 5.2.1.4 employmentType

5.2.2 Functions

- 5.2.2.1 addEmployee(name, nationalID, position, employmentType, salary)
- 5.2.2.2 removeEmployee(nationalID)
- 5.2.2.3 isExist(nationalID)

5.2.3 Related Use Cases

- 5.2.3.1 UC16: Add Employee
- 5.2.3.2 UC17: Remove Employee

5.3 HourlyEmployee

5.3.1 Attributes

- 5.3.1.1 hourlyRate

5.3.2 Functions

5.3.3 Related Use Cases

5.4 SalariedEmployee

5.4.1 Attributes

- 5.4.1.1 salary

5.4.2 Functions

5.4.3 Related Use Cases

5.5 CommissionedEmployee

5.5.1 Attributes

- 5.5.1.1 salary

5.5.2 Functions

5.5.3 Related Use Cases

5.6 PurchaseOrder

5.6.1 Attributes

5.6.1.1 orderNumber

5.6.1.2 date

5.6.1.3 productName

5.6.1.4 amount

5.6.1.5 customerBillingAddress

5.6.2 Functions

5.6.2.1 addPurchaseOrder(date, productName, amount, customerBillingAddress)

5.6.2.2 generateOrderNumber()

5.6.2.3 addPurchaseOrder(date, productName, amount, customerBillingAddress, orderNumber)

5.6.2.4 deletePurchaseOrder(orderNumber)

5.6.3 Related Use Cases

5.6.3.1 UC10: Add Sale

5.6.3.2 UC11: Delete Sale